Technical Architecture Documentation & System Implementation Specification

Project: Real Lending & Recruitment Platform

Document Type: Core Component & Features integrated

Core Service Modules & Inter-Platform Synchronization

- **Loan Operations & Credit Engine**:
  - *Mobile Client:* End-users submit loan applications, calculate interest previews, execute repayments, and receive drawdown notifications.
  - *Admin Portal:* Admins configure global APR interest tiers, process approval queues, adjust credit facilities (Payroll Facility), and audit transaction logs.

- **Payroll & HR Management:**
  - *Mobile Client:* Onboarded employers manage workforce profiles, run payroll, and track employee salary structures.
  - *Admin Portal:* Operators process corporate onboarding requests (Payroll Onboarding), set company-wide credit lines, and maintain the central employee directory.

- **Recruitment & Job Applications** (REAL Recruitment):

    - *Mobile Client:* Job seekers submit detailed professional profiles (education, skills, past employment) while employers publish job requisitions.
    - *Admin Portal:* HR teams screen candidate pools (Job Seekers), manage application review states (accepted, pending, rejected), and execute automated candidate notifications.

- **Shared Third-Party Gateways & System State:**
    - *Synchronization Layer:* Real-time integration across both interfaces for identity verification (YouVerify), payment processing (Paystack), transactional SMS (Termii), and SMTP mail alerts.

Dashboard Overview

The Dashboard serves as the operational command center for platform administrators, delivering real-time metrics across user activity, loan portfolios, and financial totals.

1. Core Modules Provides direct access to all functional management modules across the platform:

- **User & Compliance Management:** Staff, KYC, KYC Audit Trail, Support Tickets, Employees
- **Loan & Financial Operations:** Loans, Transactions, Repayment Reports, Payroll Facility
- **HR & Recruitment Operations:** Companies, Jobs, Job Seekers, Staff Hiring, Payrolls, Payroll Onboarding
- **System Settings:** Settings, Notification Emails

2. Key Metrics & Analytics Cards

Metric Card	Description & Functional Purpose
Total users	Aggregated count of registered accounts across the platform.
Active borrowers	Total users currently holding an active or ongoing loan status.

Total loans	Cumulative record of all loan applications processed by the system.
Active loans	Count of currently disbursed loans that are actively in repayment.
Overdue loans	Flagged loans past their scheduled repayment date requiring collection attention.
Outstanding	Total unpaid principal balance remaining across all active loans (includes default rate tracking).

3. Financial Summary Card Tracks platform cash flow and portfolio balance:

- **Total disbursed:** Total capital issued out to borrowers.
- **Total repaid:** Total principal and interest collected back from borrowers.
- **Outstanding:** Net remaining balance owed to the platform.

4. Quick Actions Panel Direct operational guidance directing admins to high-priority workflows:

- Reviewing pending **KYC submissions** for identity verification.
- Managing **overdue loans** and default flags.
- Updating **admin bank details** for system disbursements and collections.

Users Management Page

The Users page provides centralized user account administration, allowing administrators to monitor, filter, modify, and manage permissions for all registered platform accounts.

1. Top Control Bar

- **Search Field:** Filters users dynamically by email address, full name, or phone number.
- **Refresh Button:** Re-evaluates and updates the user table data in real time.
- **Export CSV:** Downloads the filtered or full user registry into a CSV file for auditing.
- **Delete All Users:** A high-level bulk administrative action to purge all user accounts.

2. User Data Table

Column	Description
USER	**Displays the full name and associated email address of the registered account.**
TYPE	**Account role designation badge (e.g., borrower, Admin).**
PHONE	**Primary contact phone number linked to the profile.**

CREATED	**Timestamp indicating exact profile registration date and time.**
ACTIONS	**Contextual action dropdown menu per user row.**

3. Action Menu Capabilities Clicking the action menu (three dots) on any individual user row exposes four core administrative options:

- **Edit:** Opens user profile details to update contact or account information.
- **Make Staff:** Elevates a standard user/borrower role to internal administrative staff.
- **Blacklist:** Restricts user access and blocks future platform activities/loan requests.
- **Delete:** Permanently removes the individual user account from the system.

Staff Management Page

The Staff Management module provides internal access control and role assignment capabilities, allowing super administrators to provision new staff accounts and set granular operational permissions.

1. Top Control Bar & Summary Cards

- **Search Field:** Allows real-time filtering of administrative staff members by name or email.
- **Refresh Button:** Syncs staff records with the backend database.
- **Create Staff Button:** Triggers the staff account creation modal.
- **Summary Cards:** Displays total counts for overall staff and specific assigned responsibility domains:
    - Total Staff
    - User Management
    - Loan Management
    - Payroll Management

2. Staff Data Table

Column	Description
STAFF MEMBER	Displays the full name and primary email address of the administrator.
RESPONSIBILITIES	Lists assigned access permissions and functional scope.
ADDED	Timestamp indicating account creation date.

ACTIONS	Contextual menu for editing permissions or revoking staff access.

3. Create Staff Account Modal

When creating an administrative account via the modal overlay, super admins enter basic credentials and assign responsibility roles:

- **Basic Credentials:**
  - First Name & Last Name
  - Email & Phone Number
  - Password (Optional): Can be explicitly set or left blank to auto-generate and send via email.

- **Assign Responsibilities (Role Checkboxes):**
  - **User Management:** Access to manage registered users and account states.
  - **Loan Management:** Authorization to review, approve, or decline loans.
  - **KYC Management:** Rights to review identity verification submissions.
  - **Payroll Management:** Access to payroll setup, facilities, and disbursements.
  - **Company Management:** Rights to manage employer entities.
  - **Job & Recruitment Management:** Access to job postings, applicant tracking, and staffing requisitions.
  - **Support Tickets:** Operations access for resolving user support tickets.
  - **Reports & Analytics:** Access to financial, audit, and repayment analytics.
  - **Settings:** Access to general application configurations.

KYC Dashboard & Review Page

The KYC (Know Your Customer) module enables administrative review and verification of identity documents submitted by users through the mobile application before granting access to financial and recruitment services.

1. Filter Navigation & Controls

- **Search Bar:** Filters submissions by user name or email address.
- **Refresh Button:** Re-evaluates current queue data.
- **Filter Tabs:** Categorizes user identity submissions by status:
    - Pending Review: Submissions awaiting admin evaluation.
    - Fully Verified: Approved identity profiles.
    - Unverified / Rejected: Submissions flagged as incomplete or failed.
    - All Submissions: Comprehensive record of all KYC entries.

2. Submissions Table

Column	Description
USER	Displays applicant's full name, email, and a direct "View details" link.
STATUS	Current state badge (e.g., Pending Review, Fully Verified, Rejected).
LAST UPDATED	Timestamp of last submission or state change.

ACTIONS	Direct quick-action dropdown menu (View Details, Approve, Reject).

3. Detailed KYC Review Page

Clicking **View Details** opens an in-depth validation suite organized into functional cards:

- **User Contact & Status:** Displays verification status tag, last update timestamp, phone number, and registered email address.
- **Verification Results:** Automated system check indicators for key identity parameters:
    - BVN Verification (yes/no)
    - NIN Verification (yes/no)
    - Government ID & Face match (yes/no)
    - Bank Account verification
- **Documents Section:** Displays uploaded ID type (e.g., Passport, Driver's License), document ID number, expiration date, and image previews for Government ID, Selfie, and Face Verification.
- **Facial Comparison Metrics:** Displays biometric comparison outcome (Match), along with AI confidence score percentages against required system thresholds (e.g., Confidence Level: 82.7% vs. Threshold: 70%).
- **Bank & Address Info:** Validates primary payout account details (Bank Name, Account Number) and residential address.
- **Admin Decision Menu:** Top action button allows administrators to formally **Approve** or **Reject** the applicant's identity profile.

Transactions Page

The Transactions page provides an audit trail of all financial movements occurring across the platform, including loan application fees, disbursements, and repayments.

1. Search & Filter Controls

- **Search Field:** Filters financial records using reference IDs or user details.
- **Refresh Button:** Syncs transaction history with recent payments.
- **Export CSV:** Downloads transaction records for financial auditing and reporting.
- **Type Dropdown (All types):** Filters entries by transaction classification (e.g., application_fee).
- **Status Dropdown (All statuses):** Filters entries by transaction state (e.g., successful, pending, failed).

2. Transaction Ledger Table

Column	Description
USER	**Primary identity and email address of the payer or payee.**
TYPE	**Categorization tag representing the transaction purpose (e.g., application_fee).**
AMOUNT	**Monetary value processed in the transaction (e.g., ₦1,000).**

STATUS	**State badge indicating settlement outcome (successful, pending).**
REFERENCE	**System-generated UUID string tracking payment gateway reconciliation.**
CREATED	**Timestamp recording transaction initialization.**

KYC Audit Trail Page

The KYC Audit Trail page provides an immutable compliance log tracking every identity submission, status change, and administrative action performed across user verification workflows.

1. Search & Filter Controls

- Search Input: Filters audit records by specific user name, admin email, or event type.
- Refresh Button: Syncs the audit ledger with real-time log entries.
- Event Filter (All events): Isolates specific action types (e.g., submitted, approved, rejected).
- Actor Filter (All actors): Filters entries based on who performed the action (user vs. admin).
- Date Range Pickers: Restricts log entries to a specific timeframe (mm/dd/yyyy start and end dates).

2. Audit Log Table

Column	Description
WHEN	Exact timestamp recording when the identity verification event occurred.
USER	Account owner subject to the KYC review process.
ACTOR	Initiator of the event, tagged by role (user vs. admin) and identifier email.

EVENT	Action classification badge (e.g., submitted, approved, rejected).
STATUS	State transition path mapping the verification change (e.g., pending_review → fully_verified).
REASON	Mandatory or optional justification notes logged by administrators upon rejection or override.

Repayment Reports Page

The Repayment Reports module tracks and logs loan repayment attempts, processing outcomes, and settlement methods across all active loan contracts.

1. Search & Filter Controls

- Search Field: Filters records by transaction reference, user details, or loan ID type.
- Refresh Button: Syncs table entries with live payment processor notifications.
- Status Dropdown (All statuses): Filters repayments by execution state (e.g., pending, successful, failed).
- Method Dropdown (All methods): Isolates payment channels used by borrowers (e.g., card, bank transfer, payroll deduction).
- Date Range Pickers: Limits reports to specific date windows (mm/dd/yyyy start and end dates).

Support Tickets Page

The Support Tickets module manages customer service inquiries, issue escalations, and user communication across mobile app users and administrators.

1. Search & Filter Controls

- Search Field: Filters tickets by subject title, user details, or assigned staff member.
- Refresh Button: Syncs support queue with incoming user requests.
- Status Dropdown (All statuses): Filters tickets by resolution stage (e.g., open, in-progress, resolved, closed).
- Priority Dropdown (All priorities): Isolates requests based on urgency level (e.g., low, medium, high, urgent).

2. Support Tickets Table

Column	Description
TICKET	**Displays ticket ID and subject headline.**
USER	**Identity and email address of the user submitting the inquiry.**
PRIORITY	**Severity level badge assigned to the ticket.**
STATUS	**Current resolution lifecycle state.**

ASSIGNEE	**Internal admin or staff member assigned to resolve the issue.**
LAST REPLY	**Timestamp of the most recent message or response in the thread.**

Payroll Onboarding Page

The Payroll Onboarding module manages incoming requests from employers seeking to integrate their company's HR and payroll management into the platform.

1. Onboarding Requests Table

- **Search Field:** Filters requests by employer name, company, or request status.
- **Refresh Button:** Syncs the queue with recent mobile onboarding submissions.
- **Status Dropdown (All statuses):** Isolates applications by state (e.g., requested, accepted, rejected).
- **Table Columns:**
    - **EMPLOYER:** Primary contact name and email of the employer applicant.
    - **COMPANY:** Name of the registered business entity (e.g., Codex).
    - **STATUS:** Current onboarding stage badge (e.g., requested).
    - **CREATED:** Date and time the request was submitted.
    - **DETAILS:** Link (View) navigating directly to the full application verification screen.

2. Onboarding Request Detail Page (/dashboard/payroll-onboarding/[id])

Clicking **View** opens an in-depth audit dashboard detailing employer credentials, company metrics, and geolocation data:

- **Header Controls:** Contains top action buttons allowing administrators to **Accept** or **Reject** the onboarding application, or return (Back) to the main list.
- **Status Card:** Displays current onboarding status (e.g., requested) and exact creation timestamp.
- **Summary Metrics:**
    - Employees: Count of enrolled staff submitted in the request.

    - Est. monthly gross: Estimated gross monthly payroll obligation (e.g., ₦0).
    - Provision: Allocated facility terms or status.
- **Payroll Configuration Card:** Displays requested payroll operational rules, including Payroll start/end dates, Workdays, Disbursement frequency, and expected payroll amounts.
- **Initial Request Card (Metadata & Geolocation):**
    - Location: GPS coordinates (Latitude, Longitude, Accuracy) recorded during mobile submission.
    - Company Name, Email, and Phone Number.
    - Business Address, City, State, and applicant Notes.
    - View toggle options (Formatted vs. raw JSON).
- **Company Draft Card:** Displays structured profile data provided for company profile creation.
- **Employees Draft Card:** Displays raw array data of bulk employee accounts submitted for onboarding.

Payroll Facility Page (/dashboard/payroll-facility)

The Payroll Facility module enables administrators to manage credit financing allocated to onboarded companies, allowing employers to run payroll seamless via platform-funded financing lines.

1. Facility Control Bar & Table

- **Search Field:** Filters facilities by company name, employer email, or facility status.
- **Refresh Button:** Syncs facility balances with recent disbursements and repayments.
- **Status Filter (All statuses):** Isolates accounts by active or inactive credit status.
- **Facilities Table Columns:**
    - **COMPANY:** Registered business entity alongside the associated employer contact details.
    - **STATUS:** Current operational state badge (e.g., active).
    - **OUTSTANDING:** Total remaining balance owed on the credit line.
    - **DRAWN:** Total credit volume drawn down by the company for payroll execution.
    - **REPAID:** Cumulative amount paid back by the company to liquidate facility usage.
    - **UPDATED:** Timestamp of the latest account adjustment or transaction.
    - **DETAILS:** Link (View) navigating to the facility breakdown and transaction history.

2. Detailed Facility View

Clicking **View** opens an analytical breakdown of an individual employer's credit line:

- **Header Controls:** Contains top navigation buttons allowing admins to **Adjust** facility parameters (credit limits/terms) or go **Back** to the main table.

- **Status Card:** Displays active status badge and the exact timestamp of the last configuration update.
- **Totals Card:** Highlights macro financial metrics for the credit line:
    - Outstanding: Unpaid principal balance.
    - Drawn: Capital deployed for payroll obligations.
    - Repaid: Settled obligations.
- **Transactions Ledger:** Itemized audit log recording every facility transaction, showing TYPE, AMOUNT, REFERENCE, and CREATED date.

Employees Management Page

The Employees page provides a centralized directory of all workforce members registered under onboarded companies across the platform's HR and payroll modules.

1. Search & Filter Controls

- **Search Field:** Filters employee records by employee name, user email, or company name.
- **Refresh Button:** Syncs table data with recent HR onboarding and employee profile additions.
- **Filter Dropdown (All):** Isolates employees based on specific status or organization categories.

2. Employees Data Table

Column	Description
EMPLOYEE	Full name, registered work email address, and unique system ID string (e.g., EMP-...).
COMPANY	Onboarded business entity employing the individual (e.g., Codex).
ROLE	Specific job title or designation assigned to the staff member (e.g., Research Analyst, Cloud Engineer).
SALARY	Configured monthly base compensation amount (e.g., ₦50,000).

STATUS	Current employment state badge (e.g., active).
CREATED	Timestamp recording profile creation and enrollment in the HR database.

Platform Settings Page

The Platform Settings module provides administrative controls to configure authentication methods, payment integrations, loan product parameters, notification integrations, and core system utilities.

1. Authentication & Account Details

- **User Verification Method:** Selects primary authentication route for mobile app user login and registration (Email (SMTP) vs Termii SMS).
- **Admin Bank Details:** Configures default collection bank details (Bank name, Account number, Account name) displayed to mobile app users for manual bank transfer repayments.

2. Legal & Authorization Documents

- **Bank Authorization Consent Form:** Custom .docx document uploader for global direct debit authorization forms. Displays active fallback forms and provides a file picker to upload new legal templates.
- **Privacy Policy:** Rich text input editor for modifying mobile app terms and user privacy disclosures.

3. Platform & Integration Gateway Configuration

- **Platform Parameters:** Global system values including Admin email, API URL, global loan boundaries (Min/Max loan amount, Min/Max loan tenure), default platform interest rates, and processing fee percentages.
- **Paystack Integration:** Secret key, public key, and transaction callback URL configuration for automated card billing and checkout flows.
- **YouVerify Integration:** API base URL and authorization keys for identity verification and KYC automated checks.
- **SMS Gateway (Termii):** API key and custom Termii sender ID (e.g., REALLENDING) settings for transactional SMS delivery.
- **Email Gateway (SMTP):** Mail server host connection settings (SMTP Host, Port, User, Password, From Email Address, and SSL/TLS toggle).

4. Loan Product Configurations

Includes product-specific tabs (Quick Loan, Student Loan, Salary Loan, Business Loan) allowing admins to define:

- **Product Controls:** Enable/disable toggle, custom product names, and repayment period types (Days / Months).
- **Financial Limits:** Product-specific minimum/maximum borrow amounts, minimum/maximum tenure, grace periods, processing fee percentages, and explicit allowed duration ranges.
- **Product-Specific Consent Form:** Custom legal agreement upload targeting single loan offerings.
- **APR Interest Tiers:** Variable interest rate structures based on duration windows (e.g., 1–7 days @ 18% APR; 8–14 days @ 24% APR) with controls to dynamically add or delete tiers.

5. Danger Zone -

- **Clear Database Button:** System utility to permanently purge user, borrower, and loan operational records while preserving administrative credentials and configurations.

Notification Emails Page

The Notification Emails settings module manages the broadcast list of administrative email addresses that receive system alerts, automated user notification copies, and platform updates.

1. Recipient List Management

- **Active Recipients List:** Displays configured email addresses designated to receive administrative notifications (e.g., admin@reallending.ng).
- **Delete Action:** Icon button next to each recipient allowing quick removal from the alert routing list.
- **Save Changes Button:** Persists updates made to the active recipient email list.

2. Add Recipient Form

- **Email Address Input Field:** Text input to specify new admin email addresses for alert routing (e.g., notifications@company.com).
- **Add to List Button:** Appends the entered email address to the active recipient list before saving.

Companies Page

The Companies module manages corporate profiles, registered employer details, and associated employee registries across the platform.

1. Search & Filter Controls

- **Search Field:** Filters registered organizations by name, registration number, or official email address.
- **Refresh Button:** Syncs company records and workforce counts with recent updates.
- **Verification Dropdown (All verification):** Filters businesses by verification state (e.g., verified, pending).
- **Activity Dropdown (All activity):** Isolates organizations by active or suspended account status.

2. Companies Table & Quick Actions

Column	Description
COMPANY	Registered company name alongside the corporate email address (e.g., Codex).
VERIFICATION	Compliance verification status badge (e.g., verified).
EMPLOYEES	Total count of enrolled staff registered under the company profile (e.g., 30).
CREATED	Timestamp recording company profile registration.

ACTIONS	More menu (three dots) providing quick administrative actions: **View Details**, **Deactivate**, and **Delete**.

3. Company Details Page

Navigating into a company profile displays profile metrics and the associated employee list:

- **Header & Action Buttons:** Includes top navigation (← Back to Companies), account status badges (verified, active), and management buttons to **Deactivate** or **Delete Company**.
- **Profile Summary Cards:**
  - Contact Information: Corporate email address and phone number.
  - Address: Street address, city, and state location details.
  - Business Information: Operating details such as Payroll Cycle (e.g., monthly).
  - Manager: Primary account administrator details including name, email, and phone number.
- **Employees Table:** Lists company staff members including EMPLOYEE details, POSITION, DEPARTMENT, SALARY structure, account STATUS, and HIRED date.

Job Seekers Page

The Job Seekers module manages job applications submitted by candidate profiles seeking placement through the platform's recruitment services.

1. Search & Filter Controls

- **Search Field:** Filters candidate profiles by candidate name, email address, or geographic location.
- **Refresh Button:** Syncs candidate submissions with recent mobile application activity.
- **Status Dropdown (All statuses):** Isolates applications by recruitment status (e.g., accepted, pending, rejected).
- **Sector Dropdown (All sectors):** Filters candidates by specialized industry vertical (e.g., Finance, Engineering).

2. Job Seeker Applications Table

Column	Description
APPLICANT	Candidate name, email address, and contact phone number.
LOCATION	Candidate's current state and city of residence (e.g., Bayelsa, Hsjdn).
SECTOR	Primary target industry focus (e.g., Finance).
EXPERIENCE	Cumulative years of recorded professional experience (e.g., 1 years).

EDUCATION	Highest completed academic qualification level (e.g., MSC).
STATUS	Current review status badge assigned by HR admins (e.g., accepted).
APPLIED	Precise date and timestamp when the application was logged.
ACTIONS	Direct action button (**View Details**) to inspect candidate credentials.

3. Application Detail Page (/dashboard/recruitment/[id])

Clicking **View Details** opens an exhaustive resume profile and recruitment review panel:

- **Actions Sidebar:**
  - **Send Email:** Direct trigger to compose and send messages to candidate email addresses.
  - **Delete Application:** Removes candidate records from the active recruitment database.
- **Application Metadata Sidebar:** Displays current application STATUS, APPLIED ON date, administrative reviewer email (REVIEWED BY), and review timestamp (REVIEWED ON).
- **Personal Information Card:** Details complete candidate demographics including full name, email address, phone number, location, date of birth, gender, nationality, and preferred contact method.
- **Professional Information Card:** Highlights HIGHEST QUALIFICATION, RELEVANT SKILLS, total EXPERIENCE, current EMPLOYMENT STATUS, attached CERTIFICATIONS, PROFESSIONAL MEMBERSHIPS, and LANGUAGES SPOKEN.

- **Most Recent Employment Section:** Summarizes past employment credentials including EMPLOYER, JOB TITLE, employment DURATION, and KEY RESPONSIBILITIES.

Developer Note: *Complete technical documentation for the system's code infrastructure—including full API route definitions, database schema migrations, service interfaces, environment configurations, and deployment scripts—is included directly within the root directory of the project source code zip package*

REAL PROJECT SUBSCRIPTION

WEBSITE APP

- **Fluent Forms - Annual**
  - Amount: **$79.00**
  - Status: Expired

- **Fluent CRM - Annual**
  - Amount: **$129.00**
  - Status: Expired

MOBILE APP

- **Zoho Email + SMTP - Monthly**
  - Amount: [] - £120.62
  - Subscribe Date: [Start 19 July 2026 End 02 August 2026]
  - Due Date: [02 August 2026]
  - Status: Expired
- **TERMII SMS**
  - Status: Active

FOR BOTH WEB AND MOBILE

- **YouVerify - MONTHLY**
  - Amount: [~~₦~~78,000.00]
  - Date: 3rd September, 2026
  - Status: Expired

DEVELOPMENT ACCOUNT

GITHUB - **Created**

EXPO - **Created**